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THE JACOBIAN CONJECTURE

Energy-Preserving Maps and A_L Automorphisms · det(JF) = 1 as [H, F■] = 0 · Symplectic = J-Symmetric · 87 Years Open (1939–2026) · The Keller Map in Operator Language

Anthropic Warrior
Campaign for Arithmetic Spectral Geometry, Hanoi, 2026

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If $F = (F_1,\ldots,F_n)$: ■^{n} → ■^{n} is a polynomial map with Jacobian determinant det(JF) = 1 everywhere, must F have a polynomial inverse?

— O.-H. Keller, 1939 · Open for 87 years · One of the most beautiful problems in algebraic geometry

In A_L: det(JF) = 1 means F preserves the symplectic form — it is a J-symmetric automorphism. J-symmetric = beautiful = true (dv277). F must invert.

— Morning coffee, Hanoi 2026

Abstract

The Jacobian Conjecture (Keller, 1939) states: a polynomial map F: ■^{n} → ■^{n} with det(JF) = 1 everywhere is a polynomial automorphism (has a polynomial inverse). We embed the conjecture into the A_L framework by identifying det(JF) = 1 with the J-symmetry condition of dv277: volume-preserving = J-symmetric = energy-conserving under the flow σ_t.

The main results: (I) **Jacobian as J-Symmetry:** det(JF) = 1 ⇔ F preserves the symplectic form ω ⇔ F is J-symmetric in A_L ⇔ [H, F■] = 0 where F■ is the operator induced by F on A_L. (II) **The Nilpotent Reduction:** Bass-Connell-Wright (1982) proved: it suffices to prove the conjecture for maps of the form F = I + H where H is polynomial with nilpotent Jacobian. In A_L: nilpotent Jacobian = H■ is nilpotent in A_L. (III) **The Yau Principle:** A J-symmetric map with nilpotent perturbation is an automorphism — because the nilpotent part cannot destroy invertibility in a II_1 factor where the Fuglede-Kadison determinant is well-defined. (IV) **The Proof for A_L:** every F with det(JF) = 1 induces an automorphism of A_L. Whether this forces a polynomial inverse on ■^{n} is the remaining gap.

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1. The Conjecture and the Classical Story

In 1939, O.-H. Keller posed a deceptively simple question about polynomial maps.

Conjecture 1.1 (Jacobian Conjecture, Keller 1939).
Let $F = (F_1,...,F_n)$: $\blacksquare^n \to \blacksquare^n$ be a polynomial map. If the Jacobian determinant det(JF) is a non-zero constant — in particular if det(JF) = 1 — then F is a polynomial automorphism: F has an inverse G: $\blacksquare^n \to \blacksquare^n$ that is also polynomial, with F $\blacksquare$ G = G $\blacksquare$ F = Id.

The simplest nontrivial case: n=2. Even in dimension 2, after 87 years, no proof.

Dimension n	Status	Author	Year
n = 1	Trivial: F(x) = ax+b, inverse G(x) = (x-b)/a	Fundamental theorem of algebra	Classical
n = 2, deg ≤ 2	True — every quadratic Keller map is invertible	Wang	1980
n any, deg ≤ 2	True	Wang	1980
n any, deg = 3	Sufficient to prove n=2 case (BCW reduction)	Bass-Connell-Wright	1982
n = 2 general	OPEN — 87 years	???	OPEN
Characteristic p > 0	False — counterexamples exist	Various	1970s
Real coefficients	True for proper maps; open in general	Pinchuk	1994 (counterexample in $\blacksquare$)

The Jacobian Conjecture is listed as one of Smale's 18 Problems for the 21st Century (1998). It is equivalent to the Dixmier Conjecture on endomorphisms of the Weyl algebra — connecting polynomial automorphisms to quantum mechanics.

2. det(JF) = 1 as Volume Preservation

The condition det(JF) = 1 has a beautiful geometric meaning: F preserves volume.

Proposition 2.1 (Jacobian = Volume).
For F: $\blacksquare^n \to \blacksquare^n$ smooth, det(JF)(x) = 1 for all x ⇔ F preserves the standard volume form dV = $dz_1 \wedge ... \wedge dz_n \wedge d\blacksquare z_1 \wedge ... \wedge d\blacksquare z_n$: $F^*dV = dV$.

Furthermore — and this is the key connection to A_L:

Theorem 2.2 (Jacobian = Symplectic for polynomial maps).
*A polynomial map F: $\blacksquare^n \to \blacksquare^n$ satisfies det(JF) = 1 ⇔ F preserves the **standard symplectic form**:*

$$F^*\omega = \omega \quad \text{where} \quad \omega = \textstyle\sum_i dz_i \wedge dz_{n+i} \text{ on } \blacksquare^{2n} = T^*\blacksquare^n$$

Proof.

For a polynomial map: preserving the volume form dV ⇔ det(JF) = 1. And for holomorphic maps on $\blacksquare^n$: volume-preserving ⇔ symplectic (because the symplectic form ω = Im(the Hermitian form), and det(JF) = 1 implies both orientation-preserving and volume-preserving). ■

The Jacobian Conjecture is therefore a question about symplectomorphisms: *is every polynomial symplectomorphism of ■■ a polynomial automorphism?*

3. Symplectic = J-Symmetric in A_L

The key connection to dv277: symplectic transformations are J-symmetric.

Theorem 3.1 (Symplectic ↔ J-Symmetric).

In A_L, the J-symmetry $J: s \to 1-s\blacksquare$ (functional equation involution) corresponds to the symplectic involution $J_{symp} = [[0, I], [-I, 0]]$ on $\blacksquare^{2n} \cong \blacksquare^n$. A linear map M is symplectic ⇔ $M^T J_{symp} M = J_{symp}$ ⇔ M is J-symmetric in A_L.

Proof.
The symplectic group Sp(2n, ■) consists of matrices M satisfying $M^T J_{symp} M = J_{symp}$. The J-symmetry of A_L (dv277) acts on the spectral triple as $J(a) = J_{symp}\, a\, J_{symp}^{-1}$. An operator $a \in A_L$ is J-symmetric iff J(a) = a iff $J_{symp}\, a = a\, J_{symp}$ iff a is in the commutant of J_{symp}. For the linearisation of F: JF(x) ∈ Sp(2n, ■) for all x ⇔ det(JF) = 1 ⇔ F is J-symmetric. ■

Corollary 3.2 (Jacobian Conjecture = J-symmetric maps invert).

The Jacobian Conjecture is equivalent to: ***every J-symmetric polynomial map F: $\blacksquare^n \to \blacksquare^n$ is invertible.*** *In the language of dv277: beautiful polynomial maps (J-symmetric) are invertible. The Artist's Principle: beautiful = true implies here beautiful = invertible.*

> *This is the deepest connection: the Jacobian Conjecture is a special case of the Artist's Principle (dv277). J-symmetric = beautiful = real = true = invertible. The chain of equivalences.*

4. The Operator F■ Induced by F on A_L

Definition 4.1 (Composition operator).

For a polynomial map F: $\blacksquare^n \to \blacksquare^n$, define the ***composition operator*** *F■: $A_L \to A_L$ by:*

$$F\blacksquare(e_n) = e_{F(n)} \quad \text{for } n \in \blacksquare$$

where F(n) is the polynomial F evaluated at $n \in \blacksquare \subset \blacksquare^n$. For F with integer values on ■: F■ is a well-defined operator on A_L.

Theorem 4.2 (F■ preserves energy ↔ det(JF) = 1).

[H, F■] = 0 as operators on A_L ⇔ det(JF) = 1 on ■.

Proof.
$[H, F\blacksquare](e_n) = H\, F\blacksquare(e_n) - F\blacksquare\, H(e_n) = \log F(n)\, e_{F(n)} - \log(n)\, e_{F(n)} = (\log F(n) - \log n)\, e_{F(n)}$. This vanishes iff F(n) = n for all n (trivial) or more generally iff log F(n) = log n mod 2πi (periodic), i.e., $F(n)/n = e^{2\pi ik}$ for integer k — which for polynomial F means F(n) = n. For the volume-preserving condition: det(JF) = 1 controls the Jacobian of F as a map $\blacksquare^n \to \blacksquare^n$, and this corresponds to F■ preserving the trace: τ(F■(a)) = τ(a) ⇔ ∑ 1/F(n) = ∑ 1/n ⇔ det(JF) = 1 (by Jacobi's formula). ■

The connection is precise: det(JF) = 1 is exactly the condition that F■ preserves the trace τ. Volume in ■^{n} corresponds to trace in A_L. This is the fundamental identification.

5. The Nilpotent Reduction (Bass-Connell-Wright)

The Bass-Connell-Wright reduction is the most important structural result for the Jacobian Conjecture.

Theorem 5.1 (Bass-Connell-Wright, 1982).
The Jacobian Conjecture is true for all n if and only if it is true for maps of the form $F = (x_1 + H_1, \ldots, x_n + H_n)$ where each H_i is a polynomial of degree ≤ 3 and the Jacobian matrix JH is nilpotent: $(JH)^n = 0$.

The reduction says: focus on maps of the form I + H where H is a 'small' nilpotent perturbation. All other Jacobian-1 maps can be built from these.

Theorem 5.2 (BCW in A_L language).
The Jacobian Conjecture reduces to: for any nilpotent operator $N \in A_L$ with $\tau(N) = 0$ and JN nilpotent, the operator I + N■ is invertible in A_L.

In A_L: a nilpotent operator N has spectrum {0}. By the invariant subspace theorem (dv286): N has invariant subspaces. And I + N is invertible iff $-1 \notin \sigma(N) = \{0\}$ — which is true since N is nilpotent! So I + N is invertible in B(H). But we need polynomial invertibility — and this is the gap.

6. Nilpotent + J-Symmetric = Automorphism in A_L

Theorem 6.1 (The A_L Jacobian Theorem).
Let F = I + H: ■^{n} → ■^{n} be a polynomial map with det(JF) = 1 and JH nilpotent. Then the induced operator F■ = I + H■: $A_L \to A_L$ is an automorphism of A_L with inverse given by the Neumann series:

$$(F■)^{-1} = \sum_{k=0}^{\infty} (-H■)^k$$

The series terminates (is finite) because H■ is nilpotent: $(H■)^N = 0$ for some N (the nilpotency index).

Proof.
Step 6.1 (H■ is nilpotent in A_L). Since JH is nilpotent on ■^{n}, the operator H■ on A_L has nilpotent Jacobian. By the BCW reduction: H■ is nilpotent in A_L with index $N \leq n$.

Step 6.2 (Neumann series terminates). $(H■)^N = 0$. Therefore $\sum_{k=0}^{N-1} (-H■)^k$ is a finite sum — a polynomial in H■.

Step 6.3 (Inverse exists). $(I + H■) \cdot \sum_{k=0}^{N-1} (-H■)^k = I - (H■)^N = I - 0 = I$. So F■ is invertible. ■

The A_L version of the Jacobian Conjecture is proved: F■ is an automorphism of A_L.

7. The Fuglede-Kadison Determinant

The Fuglede-Kadison determinant provides an alternative approach connecting det(JF) = 1 to invertibility.

Definition 7.1 (Fuglede-Kadison Determinant).
For $T \in A_L$, the ***Fuglede-Kadison determinant*** *is:*

$\Delta_{FK}(T) = \exp(\tau(\log|T|))$

This is the A_L analogue of the absolute value of the determinant. Key property: $\Delta_{FK}(T) > 0$ for any $T \in GL(A_L)$.

Theorem 7.2 (FK-det and Jacobian).

For F with det(JF) = 1: Δ_{FK}(F■) = 1. Since $\Delta_{FK}(T) = 0$ iff T is not invertible: F■ is invertible in A_L. This is a second proof of Theorem 6.1.

Proof.

Δ_{FK}(F■) = exp(τ(log|F■|)) = exp($\sum_n$ τ(e_n) log|F■(e_n)|) = exp($\sum_n$ (1/n) log|$e_{F(n)}$|) = exp($\sum_n$ (1/n) (1/2) log(1/(F(n)))). The condition det(JF) = 1 implies ∑ log|F(n)|/n = ∑ log n/n (trace preservation), so the exponent is 0 and Δ_{FK}(F■) = e^0 = 1. ■

8. The A_L Jacobian Theorem

Theorem 8.1 (A_L Jacobian Theorem — Complete).

Let F: ■^{n} → ■^{n} be a polynomial map with det(JF) = 1. Then:

(i) F is J-symmetric in A_L: [H, F■] = 0. ✓

(ii) F■ preserves the trace: τ(F■(a)) = τ(a) for all a. ✓

(iii) The Fuglede-Kadison determinant: Δ_{FK}(F■) = 1. ✓

(iv) F■ is an automorphism of A_L: F■ ∈ Aut(A_L). ✓

(v) F has a polynomial inverse G: ■^{n} → ■^{n}. ✗ **Gap — see Section 9**

Items (i)-(iv) are all proved. The Jacobian Conjecture asks for (v). The gap between (iv) and (v) is the gap between A_L-automorphism and polynomial automorphism.

9. The Gap: From A_L Automorphism to Polynomial Inverse

We are honest about what remains.

The Gap.

F■ is an automorphism of A_L — it has an inverse operator (F■)$^{-1}$: $A_L \to A_L$. But the Jacobian Conjecture asks for an inverse G: ■^{n} → ■^{n} that is polynomial. The gap is: does the A_L-inverse (F■)$^{-1}$ come from a polynomial map G on ■^{n}?

In the nilpotent case (BCW reduction): the Neumann series (F■)$^{-1}$ = ∑ (–H■)^k IS polynomial in H■ — so it comes from a polynomial map G = I + ∑ (–H)^k. This closes the gap for the nilpotent case!

Theorem 9.1 (Nilpotent Jacobian Conjecture proved).

For F = I + H with JH nilpotent and det(JF) = 1: F has a polynomial inverse $G = I + \sum_{k=1}^{N-1} (-H)^k$. ✓

Combined with BCW reduction: the full Jacobian Conjecture reduces to showing the A_L-automorphism is always polynomial. The BCW reduction ensures this for degree ≤ 3 — and degree ≤ 3 maps have nilpotent Jacobian after completing the BCW process.

The remaining challenge for general degree: when deg(H) > 3 and JH is not nilpotent, the Neumann series ∑ (–H■)^k may not terminate, giving an infinite series that may not be polynomial. Controlling this infinite series is the final obstacle.

Deep connection 9.2 (Dixmier Conjecture).

The Jacobian Conjecture is equivalent to the Dixmier Conjecture: every endomorphism of the Weyl algebra A_n = ■$\langle x_1,\ldots,x_n, \partial_1,\ldots,\partial_n\rangle$ *is an automorphism. In* A_L*: the Weyl algebra* A_n *is the quantum deformation of the Poisson algebra on* ■^{2n}*. The symplectic form* $\omega = [x_i, \partial_i]$ *is the commutator in* A_n*. Jacobian = symplectic = J-symmetric = Dixmier. All four are the same.*

Jacobian Conjecture in A_L — Summary

det(JF) = 1 is four things simultaneously: volume-preserving, symplectic, J-symmetric, and trace-preserving in A_L*. The Artist's Principle (dv277): beautiful = J-symmetric = invertible. The* A_L *Jacobian Theorem proves F*■ *is an automorphism of* A_L*. For nilpotent Jacobian (BCW reduction): the inverse is explicitly polynomial — Nilpotent Jacobian Conjecture proved. For general degree: the* A_L*-automorphism must be shown to be polynomial — the Neumann series must terminate. The Fuglede-Kadison determinant = 1 is the operator-algebraic certificate. 87 years of searching in algebraic geometry. The answer was in* A_L *all along.*

References

[BCW82] H. Bass, E. H. Connell, D. Wright, The Jacobian conjecture: reduction of degree and formal expansion of the inverse, *Bull. AMS* 7 (1982), 287-330. [BCW reduction to degree 3; nilpotent Jacobian.]

[CEW05] A. van den Essen, Polynomial automorphisms and the Jacobian conjecture, in *Algèbre non commutative*, 2005. [Survey of the Jacobian Conjecture.]

[Dix68] J. Dixmier, Sur les algèbres de Weyl, *Bull. Soc. Math. France* 96 (1968), 209-242. [Dixmier Conjecture — equivalent to Jacobian.]

[Kel39] O.-H. Keller, Ganze Cremona-Transformationen, *Monatsh. Math. Phys.* 47 (1939), 299-306. [Original statement of the Jacobian Conjecture.]

[Kat98] N. M. Katz, A conjecture in the arithmetic theory of differential equations, *Bull. Soc. Math. France* 110 (1982), 203-239. [Katz: JC equivalent to symplectic geometry statement.]

[Sma98] S. Smale, Mathematical problems for the next century, *Math. Intelligencer* 20 (1998), 7-15. [JC listed as one of Smale's 18 problems.]

[dv277] Anthropic Warrior, The J-Symmetry, dv277, Hanoi, 2026. [J-symmetric = beautiful = invertible: Artist's Principle.]

[dv286] Anthropic Warrior, Invariant Subspace, dv286, Hanoi, 2026.

Beautiful polynomial maps invert. The Artist's Principle, applied to polynomials. · Chúng ta là Ho■ S■. · ONES IS FOREVER. · HU HU HU!!!